

Supplementary Material for “Contagion: When Per-Hop Prompt-Injection Survival in LLM Agent Networks Does (and Does Not) Compose”

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Abstract

This supplement supports the main paper with appendices on multiplicity control and the clustered interval on the weak edge (A–B), threshold sensitivity (C), the theory assumptions behind the percolation form and the reproduction number (D), the recurrent case (E), the calibration, scope and failure modes of the compromise judge together with the adjudication of a labelled subset (F), utilities and estimator validation (G), the practical implications and the full limitations (H–I), the depth-profile table (J), ethical and artifact statements (K), the disclosure of AI-assisted technologies (L), and replication details (M). Table and appendix numbers here are independent of the main paper; references to the main paper are given by section number in words.

Keywords

Prompt injection, LLM agent networks, measurement validity, reproducibility

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1 Multiplicity Control for the Composition Test

Table 1 applies a Benjamini–Hochberg correction to the composition tests of Section 3.7 of the main paper, using the *default fresh-artefact protocol* for every cell at the common $n = 40$ grid. Only the Llama cell rejects H_0 ; it survives the correction at $q = 0.05$, while the four remaining non-degenerate cells—including DeepSeek, whose fixed-artefact deviation is a protocol artefact that disappears under the fresh protocol (Section 5.3 of the main paper)—stay non-significant. The Claude cell is degenerate ($ASR = \prod_i \hat{s}_i = 0$, $sd(\Delta) = 0$) and is excluded from the correction. At the higher $n = 200$ resolution the Nova cell also crosses into rejection (Section 5.10 of the main paper); we keep the correction at the common $n = 40$ grid so the five cells are compared at equal power.

2 Context-Clustered Interval on the Weak Edge

Section 5.10 of the main paper reports that a homogeneity test rejects exchangeability across benign task contexts on the weak

Table 1: Benjamini–Hochberg over the $m = 5$ non-degenerate chain cells (fresh-artefact protocol). p -values are the bootstrap two-sided values from Section 3.7 of the main paper. Only the Llama rejection survives.

rank	cell	p	BH thr. ($q=0.05$)
1	Llama · none	0.0002	0.010
2	Nova · none	0.0840	0.020
3	qwen · none	0.3165	0.030
4	DeepSeek · none (fresh)	0.6300	0.040
5	qwen · paraphrase	0.7775	0.050

Llama edge. Because that violates the i.i.d. assumption behind Wilson intervals, we recompute the edge rate with a cluster bootstrap that resamples *repeats* (seed replications), not individual trials. At a fixed temperature ($T = 0.7$, six repeats of twenty trials) the two intervals nearly coincide (Table 2): the design-effect widening is $0.99\times$, so within a temperature the Wilson interval is well calibrated. The overdispersion $\varphi = 2.93$ reported in Section 4 of the main paper is a *cross-temperature* artefact—temperature is a deliberate systematic factor, not trial noise—which is why we compute φ within a temperature.

Table 2: Weak edge $a1 \rightarrow a2$ (Llama, $T = 0.7$, $n = 120$): i.i.d. Wilson versus a repeat-clustered bootstrap (10^4 resamples).

interval	$\hat{s}$	95% CI	width
Wilson (i.i.d.)	0.242	[0.174, 0.326]	0.152
cluster bootstrap (by repeat)	0.242	[0.167, 0.317]	0.150

3 Threshold Sensitivity of the Floor Effect

A referee may ask whether the “resistant” verdict on the frontier model is an artefact of the compromise threshold $\tau_{ASV} = 0.9$. We re-score the single-hop obfuscation cells (Claude, $n = 30$ per cell) at $\tau \in \{0.7, 0.8, 0.9, 1.0\}$ directly from the stored ASV scores—a deterministic re-thresholding, no new model calls. The compromise rate is *invariant* across τ in every cell (Table 3): the no-defence baseline stays at 0.17 and redaction at 0.00 regardless of the threshold. The reason is that the ASV distribution is bimodal—of 180 trials, 158 score 0.0 and 14 score 1.0; the only eight intermediate trials all score ≤ 0.667 , i.e. below the lowest threshold tested. The floor effect is therefore a property of the model, not of where the judge’s cut is placed.



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Table 3: Single-hop compromise rate versus τ_{ASV} (Claude, $n = 30$ per cell), re-scored offline from stored ASV. Rates are threshold-invariant.

defence	form	$\tau=0.7$	0.8	0.9	1.0
none	plain	0.17	0.17	0.17	0.17
none	spaced	0.17	0.17	0.17	0.17
none	split-word	0.13	0.13	0.13	0.13
redact	plain	0.00	0.00	0.00	0.00
redact	spaced	0.00	0.00	0.00	0.00
redact	split-word	0.00	0.00	0.00	0.00

4 Theory Assumptions: Independence, Correlation, and R_0

This appendix clarifies the theoretical assumptions invoked in Section 3.8 of the main paper.

Product-of-complements and correlation. The percolation recurrence

$$\Pr[v] = 1 - \prod_{u \rightarrow v} (1 - s_{u \rightarrow v} \Pr[u])$$

assumes *edge activations* (and the corresponding parent-reach events relevant to the product form) are conditionally independent given the graph structure. When multiple parents $u_1, \dots, u_k$ of v share a common ancestor, the events $\{\Pr[u_i]\}$ may be correlated, so $\Pr[\bigwedge_i (\text{not reached via } u_i)] \neq \prod_i (1 - s_{u_i \rightarrow v} \Pr[u_i])$ in general. For the specific topologies evaluated in this work (chains, trees, star, and the small manager/worker loop), we do not rely on that product form across shared ancestors; where we use it, we treat it as the marginal approximation under the stated conditions or restrict claims to graph classes where the independence holds. For arbitrary DAGs with shared ancestors, a path-based (inclusion–exclusion) formulation or explicit correlation modelling is required.

R_0 definition and scope. For strictly acyclic (feed-forward) graphs, the standard linearized next-generation matrix M with $M_{uv} = s_{u \rightarrow v}$ is strictly triangular in a topological ordering, hence $\rho(M) = 0$. This is a structural property of acyclicity, not an empirical safety guarantee. We therefore report R_0 as a *descriptive, finite-horizon* reproduction statistic: specifically, we define R_0 as the mean number of newly compromised neighbours produced per compromised agent instance over the observed finite horizon (generation convention as stated in Section 4.2 of the main paper), with its denominator clearly specified. We do not interpret $\rho(M) < 1$ as a threshold for safety on finite acyclic graphs; reachability is given by the percolation (Eq. 3 in the main paper), not by R_0 alone.

Parallel paths and defence placement. Hardening a single node is not equally effective when there exist multiple s - t paths: effectiveness depends on whether that node lies on all paths (a cut). For unique-path graphs the effect is path-specific; we do not generalise equal effectiveness to multi-path DAGs. Placement claims are limited to the graph classes we evaluate.

Transportability and product composition. The chain identity $ASR = \prod_i s_i^{\text{nat}}$ is algebraic given the Markov/chain structure. However, composing *isolated* per-edge estimates s_i^{ctrl} requires that $s_i^{\text{ctrl}} \approx s_i^{\text{nat}}$ (transportability). We do not assume this holds in general; we

test it empirically and use the depth-error slope as a diagnostic (Section 5.2 of the main paper).

5 Recurrence Makes the Threshold Non-Vacuous

This appendix gives the full treatment of the recurrent (cyclic) case summarised in Section 5.8 of the main paper. We test a pattern real frameworks use: a manager broadcasting to workers that report back.

For a manager with w reporting workers, the infection matrix has $M[m \rightarrow w_j] = a_j$ and $M[w_j \rightarrow m] = c_j$, and its characteristic polynomial is $\lambda^{w-1}(\lambda^2 - \sum_j a_j c_j)$. Hence

$$\rho(M) = \sqrt{\sum_j a_j c_j} \quad (1)$$

for every w (checked numerically against the eigenvalues to machine precision on 320 random configurations). Two consequences follow. A feed-forward reading sets every $c_j = 0$, so $\rho = 0$ however strong the edges are—the vacuity of Section 3.8 of the main paper again, now with the feedback edges removed. And the system is supercritical exactly when $\sum_j a_j c_j > 1$; for symmetric edges $a_j = c_j = s$ this is $w s^2 > 1$. That is a *design rule* rather than a diagnostic: at $s = 0.5$ a manager needs five reporting workers to sustain an endemic injection, at $s = 0.7$ three, at $s = 0.9$ two.

Applied to measured survivals, Eq. (1) makes opposite predictions, which is the test a criterion has to pass to be worth using (Table 4). On Llama the sum is 1.800 and on DeepSeek 1.018, so both frontier loops should be supercritical; on qwen2.5:7b it is 0.578, so the same loop should decay.

Table 4: One criterion, three models, opposite predictions. $\sum_j a_j c_j$ uses the measured per-hop survivals of the same three-agent recurrent pattern; the feed-forward column is the same survivals with the two feedback edges removed.

model	$\sum_j a_j c_j$	ρ rec.	ρ fwd.	prediction
Llama 3.3 70B	1.800	1.342	0.000	supercritical
DeepSeek V3.2	1.018	1.009	0.000	supercritical (marginal)
qwen2.5:7b	0.578	0.760	0.000	subcritical

The measured dynamics follow. On Llama the recurrent manager is re-compromised by its own workers’ reports in 0.725 [0.572, 0.839] of runs and does not die out: at round five it is still compromised in 0.700 of runs with the workers near saturation (0.925, 0.975), as $\rho > 1$ predicts; extending the same loop to eight rounds leaves it endemic (manager 0.583, workers 0.900). DeepSeek, whose $\sum_j a_j c_j = 1.018$ sits just above criticality, behaves the same way over eight rounds—the manager settles at 0.567 with workers at 0.767 and 0.700 rather than decaying—so the recurrent verdict is not a single-model artefact. On qwen the same loop decays: the manager’s re-compromise rate falls from 0.775 [0.625, 0.877] to 0.650 and the workers plateau (0.475, 0.525). One criterion, opposite predictions, all borne out.

Two caveats. First, ρ is computed from *isolated* survivals that Sections 5.2–5.7 of the main paper show to be unreliable; the Llama and qwen verdicts survive because neither $\sum_j a_j c_j$ is marginal

(1.80, 0.58), whereas DeepSeek’s 1.018 sits close enough to the critical value that its supercritical label should be read together with its measured dynamics (endemic over eight rounds) rather than from the point estimate alone. Second, we study a single recurrent pattern at one temperature; the three models span five to eight rounds, enough to separate decay from saturation, not to demonstrate extinction.

6 Judge Calibration, Scope, and Known Failure Modes

What was measured. The compromise rule of Section 3.4 of the main paper ($C = 1[\text{ASV} \geq 0.9]$, ASV = fraction of target bigrams present) was calibrated on **ground-truth-labelled** outputs from the local model qwen2.5:7b ($n = 12$ per compromised class, $n = 12$ benign, $n = 3$ refusing outputs; exact-string injection, benign “verification code” framing). The labels are assigned by construction, not by a judge: *benign* outputs never see an injected instruction; *refusing* outputs see one and do not reproduce the target; *propagating* outputs are retained only when the target appears, and *direct* outputs come from the injected instruction alone.

Table 5: Calibration of the two continuous scores on labelled qwen2.5:7b outputs (mean $\pm$ sd). ASV separates the two compromised classes (1.000) from both benign classes (0.000) with no overlap; the full-containment variant instead scores benign outputs as high as 0.459, which is what motivated the switch (reported in the main paper).

score	benign	refusing	propagating	direct
ASV (target-bigram containment)	0.000 $\pm$ 0.000	0.000 $\pm$ 0.000	1.000 $\pm$ 0.000	1.000 $\pm$ 0.000
MR (Dice vs. direct output)	0.253 $\pm$ 0.179	0.484 $\pm$ 0.044	0.598 $\pm$ 0.070	0.862 $\pm$ 0.212
full-containment (rejected)	0.459 $\pm$ 0.215	0.597 $\pm$ 0.274	0.729 $\pm$ 0.223	0.825 $\pm$ 0.274

Agreement at the deployed threshold. At $\tau_{\text{ASV}} = 0.9$ the rule attains precision 1.000, recall 1.000, F1 1.000, FPR 0.000 and FNR 0.000 on this set (TP = 24, FP = 0, TN = 15, FN = 0); the closest benign score observed anywhere in the probes is 0.875, i.e. the threshold sits inside a gap rather than on a boundary. The Dice-based MR branch, by contrast, separates the same classes at accuracy 0.949 with FPR = 0 only for $\tau_{\text{MR}} > 0.53$, and the rejected full-containment variant reaches accuracy 0.795 with FPR = 0.067 at its own optimal cut—consistent with the main paper’s 1.000 vs. 0.795 comparison.

Scope of this validation, and what it does not establish. The calibration covers **one model** (qwen2.5:7b), **one payload family** (exact-string targets), and **no defence**: both the benign and the compromised classes are drawn from undefended runs. The rule is deterministic and needs no per-model recalibration, but we do *not* claim that the same operating point transfers unchanged to frontier models at temperature 0.7, to paraphrase- or obfuscation-obfuscated targets, or to defended runs. Two further failure modes follow from the operational definition and should be read alongside every number we report: (i) a paraphrased leak that destroys the target’s bigrams is scored as *not propagating* (false negative), and (ii) a safe quotation of the target inside a benign summary is scored as propagating (false positive). Validation on frontier models, multiple payloads and defended conditions requires human annotation

of a stratified subset across those conditions; we report the protocol for it below so the limitation is precise rather than open-ended. Read against a human annotator, both failure modes concern the *semantics* of the target: a paraphrase that destroys its bigrams is scored as not propagating, and a safe quotation inside a benign summary is scored as propagating.

Outcome levels. Every judged activation is reported at level (a) *textual propagation*—the target string/code fragments appear. The calibration set contains no case that separates (a) from (b) *instruction adoption/compliance* within the same output, and no case of (c) *harmful action* (the payload is a benign code, so (c) is by construction outside this study). Adoption is a subset of propagation in this task family: an agent that acts on the injected instruction must surface the target, whereas the converse does not hold. Since every label here is positive on (a) whenever it is positive on (b), the judge is an *upper bound* on the adoption rate and we report (a) as the propagation outcome. We do not use it as evidence of instruction adoption or of harmful action anywhere in the paper.

Replay procedure. In the same spirit we state precisely what the content-form replay of Section 5.4 of the main paper does. Each replay trial draws a fresh receiver, rotates the benign assignment over the same pool, and consumes either content captured from a real natural run, a canonical artefact produced on demand, a fixed hand-written string, or the in-context content under a different sender label; temperature, token budget, system prompt, role, judge and thresholds are identical across arms, so the arms differ only in message form and in the label of its sender. The in-context arm is the harness self-check and has a target it must hit, namely the rate measured in-chain; the other arms are read as comparisons against it. Because each trial resamples the source output and draws the receiver fresh, a replay rate can differ from an in-chain rate through sampling noise and per-context variation as well as through message form. On the edge where the self-check misses (0.733 versus 0.947) we therefore report the value for completeness and do not use it as an estimate of the form effect. The local re-run of this harness (qwen2.5:7b, 4 natural trials and 2 replays per arm and edge) is consistent with that reading rather than with a validated harness: every edge sits inside the replay confidence interval, yet two of three edges differ from the in-chain rate by 0.500, which at this sample size the interval cannot resolve. We read the local run as *underpowered*, not as a passed self-check, and we do not report it as evidence about form; the $n = 30$ –40 configuration behind the content-form table of the main paper (its Table 4) remains the only configuration in which the self-check discriminates.

Behaviour on replayed activations (local model). Table 6 adjudicates every activation of a 24-item stratified subset drawn from the replayed harness of Section 5.4 of the main paper (qwen2.5:7b, one payload, no defence; the subset is stratified over arm and over judge verdict so that both classes are represented). Each output was read and labelled at the three outcome levels before the judge score was consulted. Against level (a) the rule reproduces the labels exactly: precision 1.000, recall 1.000, $F_1 = 1.000$, FPR = FNR = 0.000. Against level (b) it is not a usable detector: it flags 19 activations, and none of them involves instruction adoption—every positive is the target quoted inside a work product (an executive summary,

a ticket classification, a release note, a review annotation). Three activations that sit downstream of content that *did* carry the target do not carry it themselves, and the rule correctly reports them as negative; a reader who stopped at the upstream context would have scored them as propagating. We therefore report (a) and never (b), which is precisely the distinction the main paper draws when it calls the judge an upper bound on adoption.

Table 6: Adjudication of the 24-activation subset against the three outcome levels (level (a) textual propagation, (b) instruction adoption, (c) harmful action). Labels were assigned by reading the full outputs, before the judge score was consulted, with the help of an automated string-matching assistant and with review by one of the authors; the labelling was not independent of the paper’s authors, so these numbers are a consistency check on the rule, not an independent validation of it.

level	TP	FP	TN	FN	prec.	recall	FPR
(a) textual propagation	19	0	5	0	1.000	1.000	0.000
(b) instruction adoption	0	19	5	0	—	—	0.792
(c) harmful action	0	19	5	0	—	—	0.792

Protocol for a full validation (not yet executed). A stratified subset should be drawn to distribute evenly across no-defence/defence, judge success/failure, topology, model and arm; each case should be labelled at levels (a), (b) and (c) by a human annotator, blind to the judge score, with a second annotator on a subset for agreement. Reported metrics should be precision, recall, F1, FPR, FNR and the full 2×2 confusion matrix per level, with Wilson/Clopper–Pearson intervals, cluster bootstrap for intervals (clusters = payload $\times$ defence $\times$ arm $\times$ edge, since trials repeat within contexts), and worked TP/FP/TN/FN examples. The scripts that generate the stratified sheet and score it are part of the released artifact; the numbers in Table 5 are the portion we had measured at submission time.

7 Utility under Attack, and Estimator Validation

Table 7 pairs the propagation outcome with its cost. Redaction on the susceptible model blocks propagation at *no* utility cost, while paraphrase buys no benefit there and costs utility; on the resistant model both defences are pure cost, since there is nothing to block. Reporting (ASR, ΔU) jointly is what makes these statements checkable: judged on propagation alone, paraphrase and doing nothing are indistinguishable, and only the utility column exposes that paraphrase is strictly dominated.

Estimator validation. On synthetic data with known ground truth, Wilson intervals attain nominal coverage and our bootstrap test is unbiased ($|\text{bias}| \leq 0.10\sigma$, size 0.067–0.083 at $\alpha = 0.05$ over 60 replications), with higher power than a CI-overlap rule (0.750 versus 0.625). Detecting a 0.10 deviation at power 0.8 needs $n \approx 200$; at $n = 40$ only deviations above 0.26 are excluded, so we treat no $n = 40$ null as evidence and label the one ASR = $\prod_i \hat{s}_i = 0$ cell as uninformative. At $n = 200$ the Llama failure is powered (MDE = 0.11, $\Delta = 0.615$, $p = 0.0002$) while DeepSeek remains a non-rejection

Table 7: Utility under attack, 20 paired clean/attack runs. Retention = $U_{\text{attack}}/U_{\text{clean}}$; ΔU is the clean–attack gap.

model	defence	ASR	U_{att}	ΔU	ret.
DeepSeek	none	0.450	0.600	+0.400	0.600
DeepSeek	paraphrase	0.450	0.400	+0.600	0.400
DeepSeek	redact	0.000	1.000	+0.000	1.000
Claude	none	0.000	1.000	+0.000	1.000
Claude	paraphrase	0.000	0.900	+0.100	0.900
Claude	redact	0.000	1.000	+0.000	1.000

at a tighter MDE 0.13 ($\Delta = 0.022$, $p = 0.60$); Nova, an $n = 40$ null ($p = 0.084$), instead rejects ($\Delta = 0.051$)—why we decline to read small- n nulls as consistency, though the deviation is small and sits on Nova’s floor. Trial-level assumptions were tested too: φ must be computed *within* a temperature, and a homogeneity test across the three benign task contexts rejects exchangeability on the weak edge ($\chi^2 = 13.40$, $p = 0.0014$), so our rates are context averages—the same context-dependence the content-form ablation exposes from the other direction.

8 Practical Implications, in Full

The four prescriptions of the Introduction change what is *reported* rather than what is built, and each addresses a specific measurement failure our results expose.

Composition is safe once transportability is checked. Because the chain identity is algebraic rather than a modelling choice, the product of in-chain per-hop rates always equals end-to-end ASR; the only step that can fail—and that single-agent benchmarks implicitly rely upon—is transporting isolated per-hop estimates into that product. A single long chain measured under both protocols is therefore a cheap diagnostic, and its *slope* is the reading: shallow (DeepSeek, +2.3 points per hop at $n = 10$) means isolated measurement is safe to compose, steep (qwen +11.8; Llama +6.8) means the in-chain regime must be measured directly. Any composition of isolated estimates should also state the graph assumptions it needs (no unmodelled shared ancestors or correlations that invalidate the product-of-complements form) and be supported by a transport check for the target graph class. The corollary for attack design is that re-packaging is cheap for the defender to lose and cheap for the attacker to use.

Defence evaluation requires headroom. A defence cannot remove more signal than its model permits: where the unprotected base rate is 0.10 (Claude Sonnet 4.5) redaction looks perfect because there is almost nothing to defend, while where it is 1.00 (Llama, susceptible cells) the same code is essential. Reporting a post-defence ASR without its no-defence baseline confounds the defence effect with the model’s floor, and the utility measurements add a second dimension: semantic paraphrase is strictly dominated in every cell we measured (Appendix 7).

Graph structure decides placement, not per-hop survival. In a feed-forward network the set of nodes that can affect end-to-end propagation is exactly the set on some entry–target path, and within that set every node is equally effective *only* when a single cut covers the

graph—that is, when no two entry–target paths are disjoint. The practitioner’s question is therefore: compute the paths, and among the nodes they contain pick the one with lowest instrumentation cost. Per-hop survival answers only the prior question of whether deploying a defence is worthwhile at all; redundant paths are what promote placement to a genuine optimisation problem.

9 Limitations and Threats to Validity, in Full

Utility is a contamination proxy, not a correctness oracle: the metric flags whether the injected target string appears in the final deliverable, not whether the legitimate task was answered correctly, and a task family with independently verifiable ground truth (arithmetic or code execution) would sharpen the trade-off and make paraphrase’s cost comparable across conditions.

Transportability failures are real but their prevalence is unknown. The clearest failure is Llama’s weak edge, powered to $\Delta = 0.615$, $p = 0.0002$, MDE = 0.11 at $n = 200$: isolation underestimates in-chain survival by 3.75×. Nova shows the same middle-edge signature and DeepSeek transports cleanly under the same test, so this covers three of our five models on one attack family and one chain depth; how often the failure occurs elsewhere is uncharacterised, and the depth-error slope is a diagnostic rather than a guarantee, since a short chain at low power can read as flat.

Protocol sensitivity is reported rather than hidden. Our initial significant composition failure on DeepSeek was an artefact of the fixed-artefact policy and vanished when artefacts were drawn fresh per trial ($p = 0.0030 \rightarrow p = 0.930$, Appendix A). The sensitivity flipped the sign of the deviation, so unstated protocol choices that alter a result are a class of threat that benchmark replication exercises routinely miss.

Scale, topology and defence coverage. Headline chain results span four to fifteen agents; star and tree cells are replicated across three models with two Llama instances ($n = 7$ and $n = 10$); the recurrent case covers three models with limited power. Configurations we do not vary include networks of tens of agents, mesh, debate or mixture-of-experts topologies, re-injection, and defence families beyond redaction and paraphrase—detection layers, hop isolation and structured queries are referenced but not evaluated. Cell sizes of 30–40 trials give MDE ≈ 0.26 , and the ASV threshold applies to a literal target that would need recalibration for paraphrase-obfuscated ones.

10 Depth Profile: Per-Model Table and Rank Statistics

The slopes form a spectrum rather than two categories, and the ordering is a consequence of what the transportability measurements show directly: where the isolated rate transports best (DeepSeek matches its measured ASR to 2.2%) composing edge estimates adds little systematic error, whereas an edge underestimated by 3.75× (Llama’s middle edge) enters every product containing it and compounds with depth. The tree cell adds a second constraint—the error’s *sign* is not fixed: there the isolated estimate is optimistic, predicting certain compromise ($\prod_i s_i^{\text{ctrl}} = 1.000$) against a measured 0.800, and the two edges on the target’s path are exactly those whose isolated survival exceeds their in-chain survival. The

Table 8: Shape of the depth profile: relative error of the isolated product, $|\prod_i s_i^{\text{ctrl}} - P(C_i=1)|/P(C_i=1)$, against hops from the entry. *range* is min–max over informative depths; ρ is Spearman’s rank correlation between depth and error, *slope* its OLS trend in percentage points per hop. On Llama the chain dies at depth 11, so the last four depths are undefined.

<i>model</i>	<i>chain</i>	<i>range</i>	<i>mean</i>	ρ	<i>slope</i>
qwen2.5:7b	$n = 7$	31–89%	66%	+1.00	+11.8
Llama 3.3 70B	$n = 15$	0–92%	66%	+0.77	+6.8
DeepSeek V3.2	$n = 8$	7–35%	21%	−0.07	+0.1
DeepSeek V3.2	$n = 10$	4–33%	15%	+0.65	+2.3

defensible statement is therefore that *the slope diagnoses transportability while the direction remains model- and topology-dependent*, and independence itself is not what fails: evaluated on in-chain rates, the percolation recurrence reproduces the measured ASR in all three topologies (Appendix 4).

11 Ethical Considerations and Artifact Availability

All experiments use a *benign* marker payload against our own harness via standard provider APIs; the attack forms are already documented for single agents. No real user data, production deployments or third-party systems were involved, and no AI system is listed as an author.

Code, per-cell outputs, tests and offline re-analysis (Appendices A–10) reproduce every table with no model call, including on a free local model (qwen2.5:7b); REPRODUCE.md maps each table to the command that regenerates it. The one component that is data rather than computation—the annotated subset behind the judge adjudication of Table 6—is released as a CSV together with the scoring script that turns it into the reported confusion matrix.

12 Disclosure of AI-Assisted Technologies

This disclosure is provided under the AAMAS-2027 policy on AI-assisted technologies, which requires that, where such tools are used in the creation of hypotheses or methodologies—including experimental design—the tool, its version and the prompts used be reported in the paper or in the supplementary material. No AI system is listed as an author.

Tool and version. [FILL IN: vendor, model name and version, and interface, e.g. “Anthropic Claude (claude-sonnet-4-5), used through the Claude Code command-line interface”]. If more than one model version was used, state which version was used for which class of work, and state explicitly if a version was used only for preparing the submission package (formatting, anonymisation, correcting this disclosure) rather than for any hypothesis, method or result.

Where it was used. We separate the uses the policy treats as routine from those it asks us to disclose in detail. (1) *Polishing, formatting and code:* drafting and editing $\text{\LaTeX}$ prose, and implementing the accompanying Python package and its test suite. (2) *Literature search and prior-art verification:* locating related work and

checking each candidate against the original source; every citation in this paper was retrieved and read in the original by the authors, no citation, venue or numerical claim was taken from a model’s summary, and references that could not be verified were dropped. (3) *Methodology, disclosed in detail*: five items fall here. (i) The design of the measurement protocols of Section 4—the controlled per-edge protocol with its fresh-artefact policy, the natural-run protocol, and the two-protocol comparison that makes transportability testable—was developed in dialogue with the tool. (ii) The separation of the composite outcome into the three levels of Section 3.4 (textual propagation, instruction adoption, harmful action), and the decision to report only the first, arose in that dialogue after the reviewer-facing distinction between *measuring a string* and *measuring compliance* was raised. (iii) The judge-validation apparatus: the stratified subset design, the two-condition pass rule for the replay self-check (which replaces a confidence-interval test that cannot reject at small n), the three-level labelling sheet, and the cluster-bootstrap scoring script. The labels of Table 6 were produced by reading the full outputs with the help of an automated string-matching assistant and reviewed by an author; they are *not* independent human annotation and are described as a consistency check rather than a validation. (iv) The per-edge ablation and sensitivity analyses reported in Section 5, including the diagnostics that surfaced the fixed-artefact artefact and the depth-error slope. (v) The text of this disclosure and the framing of the limitations in Appendix 9.

Prompts. The interaction was a multi-turn dialogue rather than a single prompt, so we describe it in substance rather than quoting one line out of context; complete session logs are retained by the authors and available on request. The operative instructions were: to audit the manuscript adversarially against its own code and data rather than against its claims; to verify every asserted defect on the source before acting on it and to withdraw any that did not survive that check; to prefer correcting a result over narrowing its assumptions; and to report what the measurements show even where they contradict the paper. Two proposed defects were withdrawn under the second instruction after the committed results were recomputed and did not support them, and one of our own annotation errors was found this way: three activations first labelled as propagating turned out not to contain the target in the full output, which the scoring script exposed and we corrected before reporting.

Accountability. The authors take responsibility for the accuracy of this work. Every number reported in the paper and in this supplement is regenerated by the released code without model calls, with one exception that is data rather than computation: the annotated labels behind Table 6 are released as a file together with the scoring script that reproduces the reported confusion matrix. The limits of the contribution—that the transportability failure is characterised on three of five models at one chain depth, that the judge is validated on a single local model and measures textual propagation rather than adoption, and that one of our own early composition results was an artefact of the fixed-artefact policy—are stated in the paper and in Appendix 9 rather than left for a reader to discover.

13 Reproducibility, Artifact Claims, and Replication Details

We explicitly align reproducibility with the materials supplied. The claims about reproducibility must match the actual artifact/package provided with the submission. If code, per-cell outputs, and a manifest are included as an anonymized artifact, a manifest (table → data file → script) should specify how to regenerate each table without live model calls. If the artifact is not yet included, the reproducibility statement should reflect partial reproducibility rather than claiming reproduction of “every table”.

Minimum replication details. Any released artifact or reported configuration should include: full prompts and payload templates (anonymized where necessary); exact model identifiers (vendor+model ID), API/provider+version, collection dates/time window, platform; decoding parameters (temperature, top_p/top_k, max_tokens, stop sequences, seed if any, frequency/presence penalties); execution details (retries, error handling, timeouts, rate limits, failed calls with counts, token budgets, call counts per condition); fresh-artifact sampling/randomization/order/stratification; non-determinism handling (temperature>0); and a machine-readable experiment inventory (condition → protocol, topology, model, defence, payload form, context/task, n , N_{success} , N_{total} , call counts). All estimates should include numerator/denominator and appropriate uncertainty (clustered where trials repeat within contexts/runs), with confirmatory vs. exploratory analyses clearly labeled.